ELEC2134 : Circuits and Signals

PART A (Week 1 to Week 5)

Transform Methods

- Periodic signals and Fourier series
- Aperiodic signals and Fourier transforms
- Laplace transforms of signals and circuits

System Analysis

- Linear systems and convolution
- Impulse response
- Frequency response and transfer functions
- Steady-state analysis of linear systems
- Transient analysis of 1st and 2nd systems
- Circuit analysis using transform methods

PART B (Week 7 to Week 10)

AC Circuit Analysis

- AC circuits with sinusoidal inputs
- Use of phasors and complex impedance in AC circuit analysis
- AC power: real, reactive and apparent power
- Power factor (leading/lagging)
- Series and parallel resonance
- Transformers and coupled coils
- Application of network theorems: Two port networks

Adapted by Aous Naman (PhD) from lectures by
Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

ELEC2134 : Circuits and Signals (PART B)

Topic 6b: AC Circuit Analysis

AC Circuit Theorems

- ✓ Impedance (Z), Admittance (Y), Conductance (G) and Susceptance (B)
- ✓ Delta(Δ) - Wye (Y) Transformation in AC networks
- ✓ Independent and dependent current and voltage sources
- ✓ Nodal Analysis
- ✓ Mesh Analysis
- ✓ Superposition Theorem
- ✓ Thevenin's and Norton's Equivalent Circuits

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Impedance, Admittance and Susceptance

- The reciprocal of resistance is known as conductance (G) and is expressed in siemens (S) or Ω^{-1}. {note this is only true if the reactance is zero, i.e resistive circuits only}

$$G = \frac{1}{R}$$

The conductance (G) of a resistor with a resistance of 5Ω is equal to 200 mS (i.e $G = \frac{1}{5}$)

- At times it is convenient to work with the reciprocal of impedance (Z) known as Admittance (Y)

$$Y = \frac{1}{Z}$$

Y is the admittance, measured in siemens (S) or Ω^{-1}

Z is the impedance, measured in ohms (Ω)

- **Admittance** is used in parallel circuit analysis

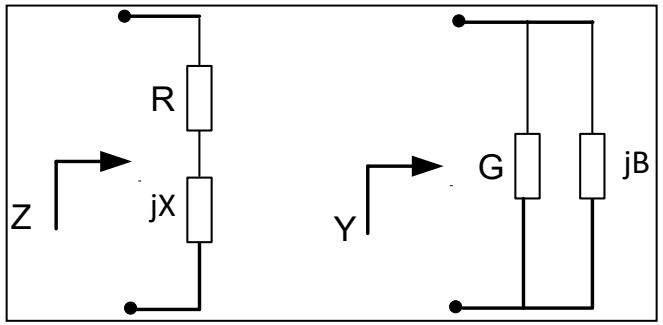
- The admittance Y can be written as

$$Y = \frac{1}{Z} = G + jB$$

- The **imaginary part** of admittance is called **susceptance** and is denoted by B and measured in siemens(S) or Ω^{-1}.

- The two components of the susceptance are capacitive susceptance B_C and the inductive susceptance B_L.

Relationships between R, X, G and B



A simple series and parallel equivalents

$$Y = \frac{1}{Z} = \frac{1}{R+jX} = \frac{1}{R+jX}\frac{R-jX}{R-jX} = \frac{R}{R^2+X^2} - j\frac{X}{R^2+X^2} = G + jB$$

- $G = \frac{R}{R^2+X^2}$ and $B = \frac{-X}{R^2+X^2}$ {G = 1/R only if X =0}
- Conversely, once G and B are known, R and X can be found as

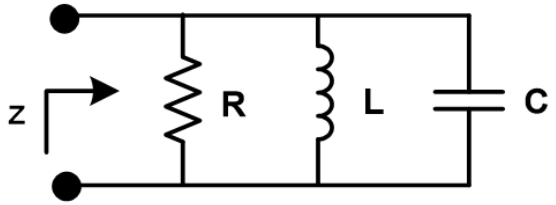
$$R = \frac{G}{G^2+B^2} \text{ and } X = \frac{-B}{G^2+B^2}$$

Basic Elements : Impedance and Admittance

Element	Impedance	Admittance
R	$Z_R = R$	$Y_R = \frac{1}{R}$
L	$Z_L = j\omega L$	$Y_L = \frac{1}{j\omega L}$
C	$Z_C = \frac{1}{j\omega C}$	$Y_C = j\omega C$

Examples

- Consider the following parallel circuit and compute the impedance Z.



Method 1: Z = R// $j\omega L$ // (1/ $j\omega c$)

Method 1: Since this is a parallel circuit, it is easier to compute the admittance(Y), first:

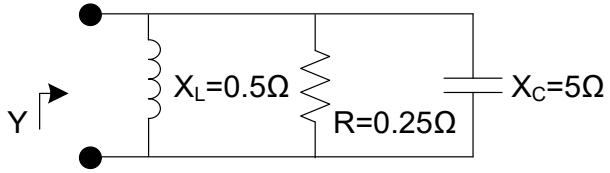
$$Y = \frac{1}{z} = \frac{1}{R} + \frac{1}{jX_L} + \frac{1}{1/jXC} = \frac{1}{R} + \frac{1}{j\omega L} + j\omega c$$

$$= \frac{1}{R} + j\left\{\omega c - \frac{1}{\omega L}\right\}$$

The reciprocal of admittance yields impedance. i.e $Z = \frac{1}{Y}$

Example 14

- Compute the impedance (Z) and Y(admittance) for the circuit shown below:



Method 1: Z = 0.25// j0.5 //(1/j5)

Method 2: Compute the admittance, first:

$$Y = \frac{1}{R} + \frac{1}{jX_L} + \frac{1}{1/jXC} = \frac{1}{0.25} + \frac{1}{j0.5} + j5$$

$$= 4 - 2j + j5 = 4 + j\,3 = 5\angle 36.9^{o}\ \Omega^{-1}$$

Since the impedance is the reciprocal of admittance

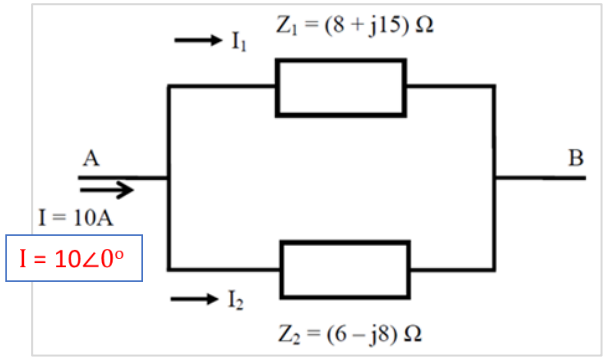
$$Z = \frac{1}{Y} = \frac{1}{5\angle 36.9^{o}} = 0.2\ \angle -36.9^{o}\ \Omega$$

Z = 0.16 – j 0.12

Examples

Example 15

- For the parallel circuit shown below, find the branch currents (I_1 and I_2), and also the supply voltage. Also draw the phasor diagram. You may assume that the phase angle of the supply current I is zero degrees.



$Z_1 = 8 + j15 = 17\angle 61.93^o$

$Y_1 = \frac{1}{Z_1} = \frac{1}{17\angle 61.93^o} = 0.05882\angle -61.93^o$

$= [27.68 - j51.9]\ 10^{-3}\ \Omega^{-1}$

$Z_2 = 6 - j8 = 10\angle -53.13^o$

$Y_2 = \frac{1}{Z_2} = \frac{1}{10\angle -53.13^o} = 0.1\angle 53.13^o$

$= [60 + j80]\ 10^{-3}\ \Omega^{-1}$

Solution:

Total admittance $Y = Y_1 + Y_2 = [87.68 + j28.1]\ 10^{-3} = 92.07\ 10^{-3}\angle 17.77^o$

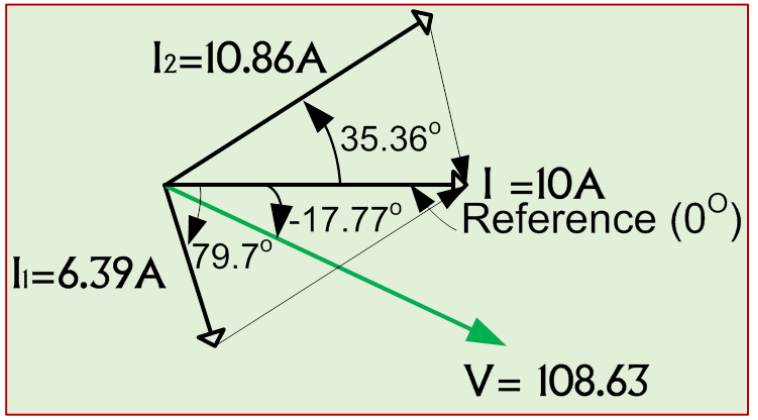
Total Impedance $Z = \frac{1}{Y} = \frac{1}{92.07\ 10^{-3}\angle 17.77^o} = 10.86\ \angle -17.77^o\ \Omega$

Supply voltage $V = I\,Z = (10\ \angle 0^o)(10.86\ \angle -17.77^o) = 108.6\ \angle -17.77^o$ Volts

The branch current $I_1 = \frac{V}{Z_1} = \frac{108.6\ \angle -17.77^o}{17\angle 61.93^o} = 6.39\angle -79.7^o$ A

The branch current $I_2 = \frac{V}{Z_2} = \frac{108.6\ \angle -17.77^o}{10\angle -53.13^o} = 10.86\angle 35.36^o$ A

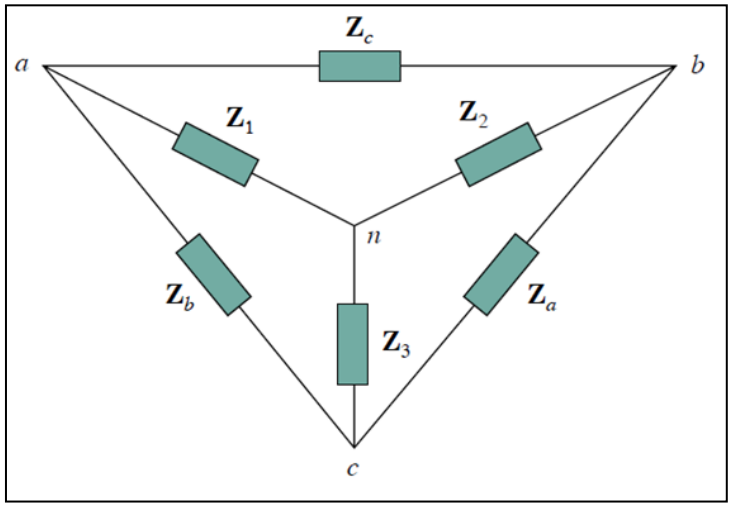
Supply Current is the reference

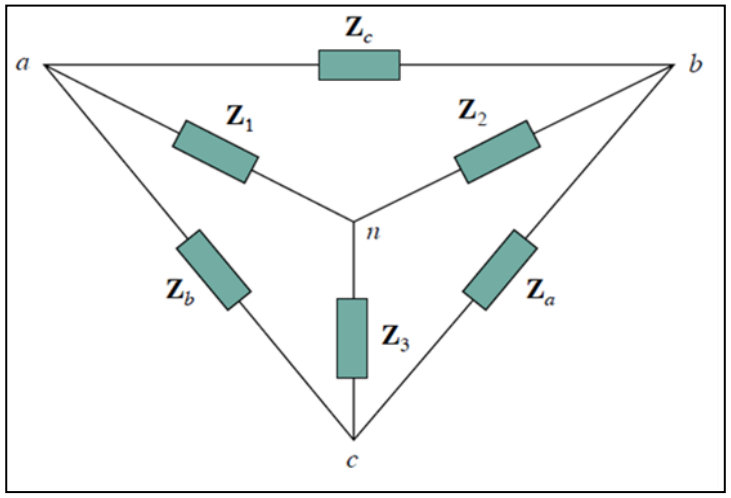


Phasor diagram

Delta(Δ) - Wye (Y) Transformation in AC networks

- A **delta** network can be converted easily to a **Wye** network (or **Star** network) or vice-versa.
- The concept of this conversion is normally used in three-phase ac circuits.





Delta(Δ) to Wye (Y) Transformation

$$Z_1 = \frac{Z_b Z_c}{Z_a + Z_b + Z_c}$$

$$Z_2 = \frac{Z_c Z_a}{Z_a + Z_b + Z_c}$$

$$Z_3 = \frac{Z_a Z_b}{Z_a + Z_b + Z_c}$$

Each impedance in the Wye network is the product of the impedance in the two adjacent Δ branches, divided by the sum of the three Δ impedances

Wye (Y) to Delta(Δ) Transformation

$$Z_a = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_1}$$

$$Z_b = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_2}$$

$$Z_c = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_3}$$

Each impedance in the Δ network is the sum of all possible products of Y impedances taken two at a time, divided by the opposite Y impedance.

Balanced Delta (Δ) or Wye (Y) circuit

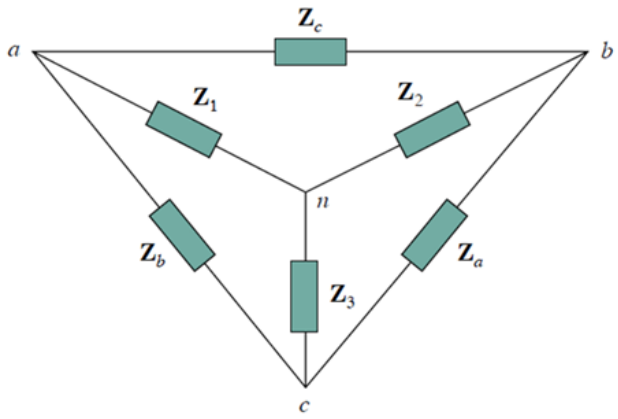
✓ A Wye or Delta circuit is balanced if it has equal impedances (load) in all three branches. That is,

$$Z_Y = Z_1 = Z_2 = Z_3$$

$$Z_1 = \frac{Z_b Z_c}{Z_a + Z_b + Z_c}$$

For a balanced load

$$Z_Y = \frac{1}{3} Z_\Delta$$



$$Z_\Delta = Z_a = Z_b = Z_c$$

$$Z_a = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_1}$$

For a balanced load

$$Z_\Delta = 3 Z_Y$$

For a balanced load the equivalent star or Wye connected impedance is 1/3 times that of the delta connected impedance.

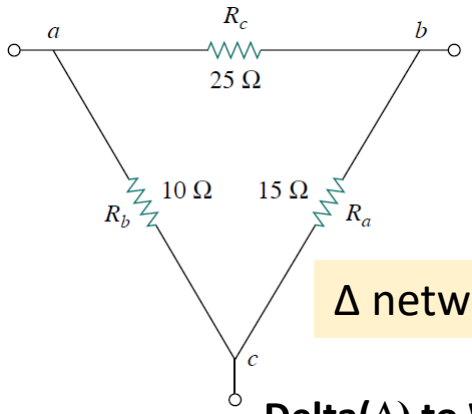
For a balanced load the equivalent delta connected impedance is 3 times that of the Wye or star connected impedance.

✓

Balanced Delta (Δ) or Wye (Y) circuit

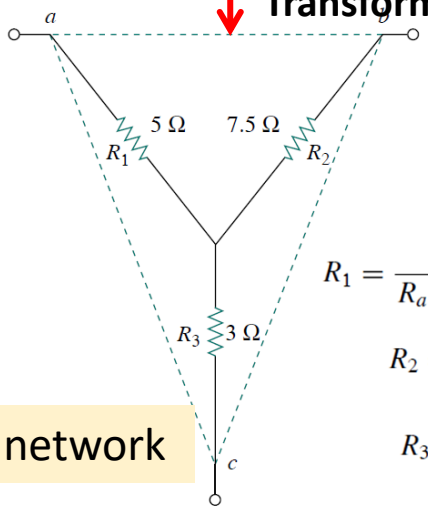
Example 16

- Convert the Δ network of resistors shown below to a Y network of resistors:



Δ network

Delta(Δ) to Wye (Y) Transformation



Y network

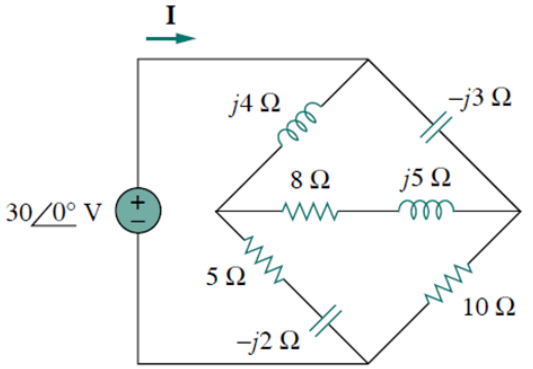
$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c} = \frac{25 \times 10}{25 + 10 + 15} = \frac{250}{50} = 5\ \Omega$$

$$R_2 = \frac{R_c R_a}{R_a + R_b + R_c} = \frac{25 \times 15}{50} = 7.5\ \Omega$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c} = \frac{15 \times 10}{50} = 3\ \Omega$$

Example 17

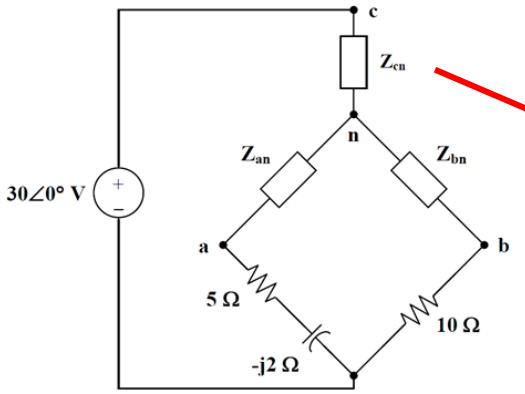
- Find the current I for the circuit shown:



$$\mathbf{Z}_{an} = \frac{j4(8+j5)}{j4+8+j5-j3} = \frac{4(-5+j8)}{8+j6} = 0.32 + j3.76$$

$$\mathbf{Z}_{bn} = \frac{-j3(8+j5)}{8+j6} = \frac{3(5-j8)(8-j6)}{100} = -0.24 - j2.82$$

$$\mathbf{Z}_{cn} = \frac{j4(-j3)}{8+j6} = \frac{12(8-j6)}{100} = 0.96 - j0.72$$



The total impedance from the source terminals is

$$\mathbf{Z} = \mathbf{Z}_{cn} + (\mathbf{Z}_{an} + 5 - j2) \parallel (\mathbf{Z}_{bn} + 10)$$

$$\mathbf{Z} = \mathbf{Z}_{cn} + (5.32 + j1.76) \parallel (9.76 - j2.82)$$

$$\mathbf{Z} = \mathbf{Z}_{cn} + \frac{(5.32 + j1.76)(9.76 - j2.82)}{(5.32 + j1.76) + (9.76 - j2.82)}$$

$$\mathbf{Z} = 0.96 - j0.72 + 3.744 + j0.4074$$

$$\mathbf{Z} = 4.704 - j0.3126 = 4.714\angle -3.802°$$

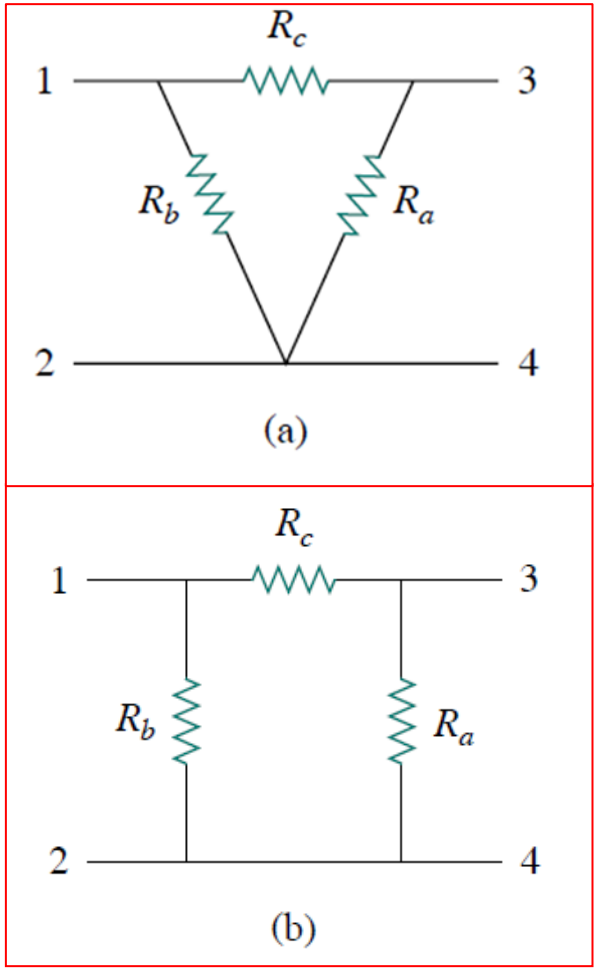
Therefore,

$$\mathbf{I} = \mathbf{V}/\mathbf{Z} = \frac{30\angle 0°}{4.714\angle -3.802°}$$

$$\mathbf{I} = \underline{\mathbf{6.364\angle 3.802°\ A}}$$

Alternative forms of Y and Δ networks

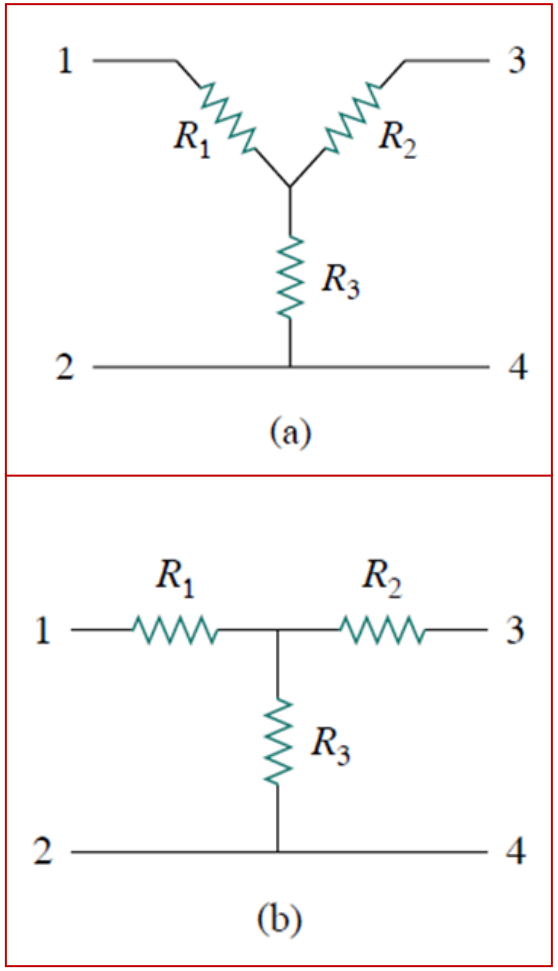
Δ-network



Pi - network

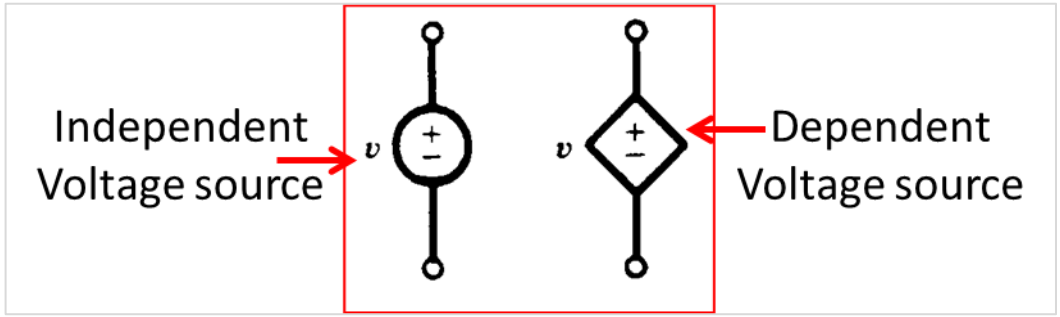
PI and T networks are commonly used in power transmission line models.

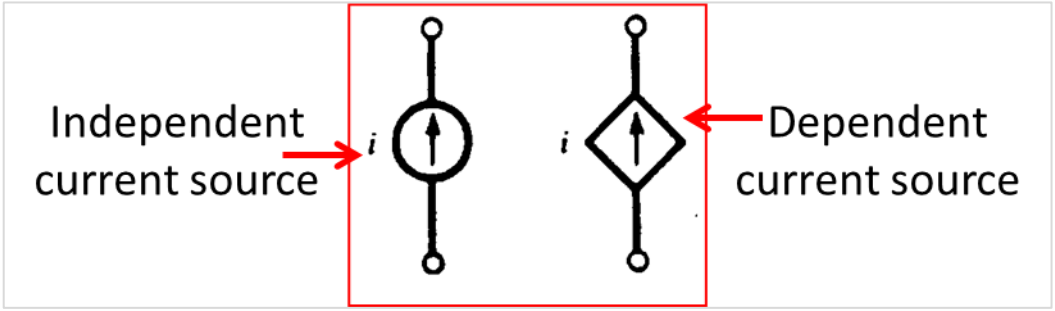
Y -network



T- network

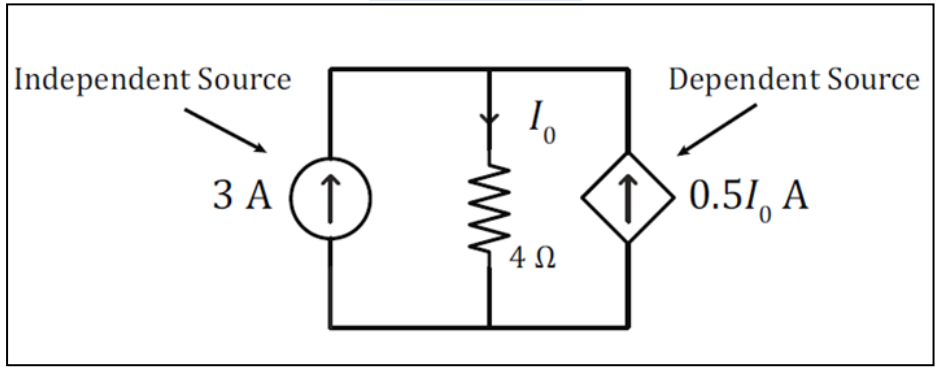
Independent and dependent current and voltage sources





Dependent source are drawn with a diamond outline

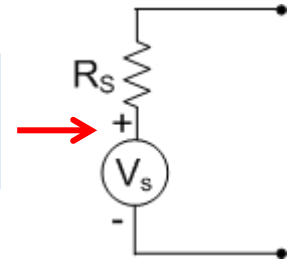
Example

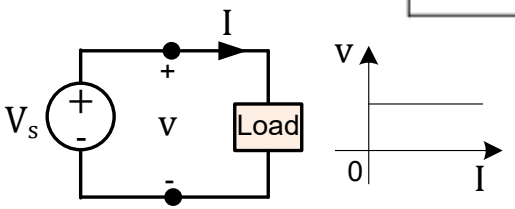


- ✓ Dependent source is controlled by another voltage or current.
- ✓ The dependent current source supplies 0.5 times the current flowing through the 4ohm resistor in the above example

Current and Voltage sources

Voltage source

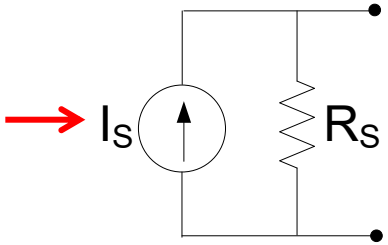


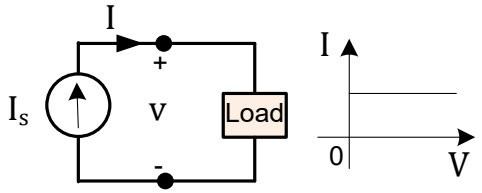


Ideal voltage source and its I-V characteristic

- ✓ An ideal voltage source has zero internal resistance (Rs = 0) and it cannot exist in nature
- ✓ A practical voltage source has a very small internal resistance .This resistance (Rs) is typically less than 1 ohm(A car battery has an internal resistance of less the 0.1 ohm)
- ✓ The internal resistance of a voltage source appears in **series** with an ideal voltage source(V_s)

Current source





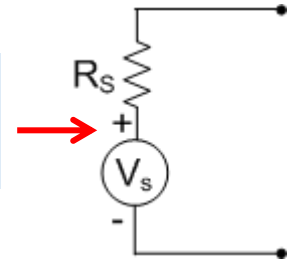
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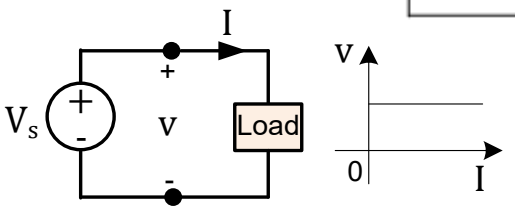
- ✓ An ideal current source has an infinite internal resistance (Rs = ∞) and it cannot exist in nature
- ✓ A practical current source has a very high internal resistance(Rs) . This resistance is typically in the order of a Mega ohm.
- ✓ The internal resistance (Rs) of a practical current source is always shown in **parallel** with an ideal current source (I_s)

Note: The current sources are rarely built with a battery and a resistor combination. Instead, transistors are connected in such a way as to produce current sources.

Current and Voltage sources

Voltage source

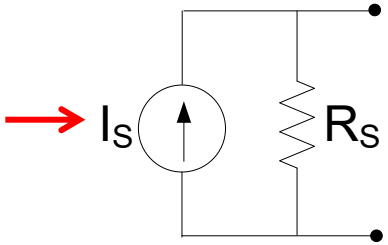


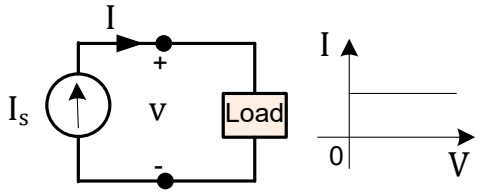


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Current source





Ideal current source and its I-V characteristic

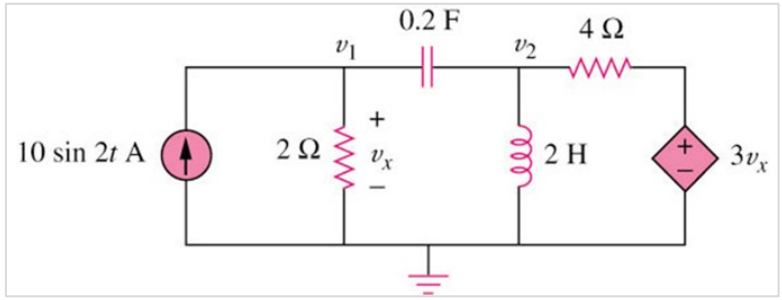
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Note: The current sources are rarely built with a battery and a resistor combination. Instead, transistors are connected in such a way as to produce current sources.

Node Voltage Analysis

Example 18

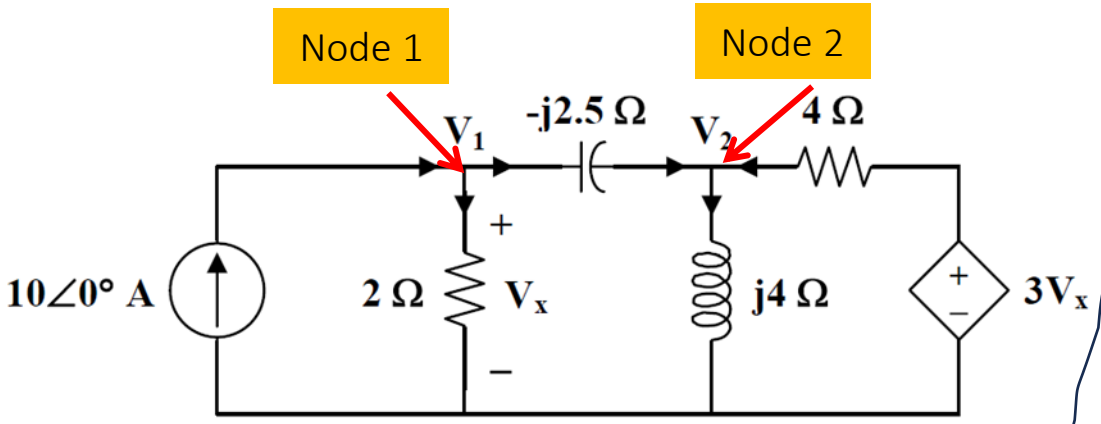
❑ Use node voltage analysis to find the voltages v_1 and v_2 in the circuit shown below:



Solution:

- Convert to Phasor Domain

$$10\sin(2t) \longrightarrow 10\angle 0°, \quad \omega = 2$$
$$2\text{ H} \longrightarrow j\omega L = j4$$
$$0.2\text{ F} \longrightarrow \frac{1}{j\omega C} = -j2.5$$



Nodal analysis uses KCL to find node voltages in a circuit.

Nodal Analysis at Node 1 provides:

$$10 = \frac{\mathbf{V}_1}{2} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{-j2.5}$$
$$100 = (5 + j4)\mathbf{V}_1 - j4\mathbf{V}_2 \quad (1)$$

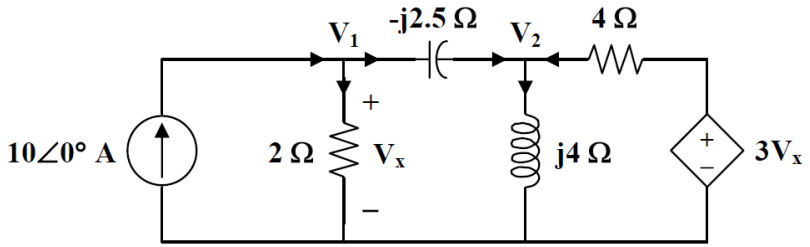
Nodal Analysis at Node 2 provides:

$$\frac{\mathbf{V}_2}{j4} = \frac{\mathbf{V}_1 - \mathbf{V}_2}{-j2.5} + \frac{3\mathbf{V}_x - \mathbf{V}_2}{4} \qquad \mathbf{V}_x = \mathbf{V}_1$$
$$0 = -(7.5 + j4)\mathbf{V}_1 + (2.5 + j1.5)\mathbf{V}_2 \quad (2)$$

Node Voltage Analysis

We have to solve following two simultaneous equations to find $\mathbf{V}_1$ and $\mathbf{V}_2$

$$\begin{bmatrix} 5+j4 & -j4 \\ -(7.5+j4) & 2.5+j1.5 \end{bmatrix}\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix}=\begin{bmatrix} 100 \\ 0 \end{bmatrix}$$



✓ Determinant of the matrix is:

$$\Delta = (5+j4)(2.5+j.15)-(-j4)(-(7.5+j4)) = 22.5-j12.5 = 25.74\angle -29.05°$$

✓ Using Cramer's rule:

$$\mathbf{V}_1 = \frac{\begin{bmatrix} 100 & -j4 \\ 0 & (2.5+j1.5) \end{bmatrix}}{22.5-j12.5} \qquad \mathbf{V}_2 = \frac{\begin{bmatrix} 5+j4 & 100 \\ -(7.5+j4) & 0 \end{bmatrix}}{22.5-j12.5}$$

$$\mathbf{V}_1 = \frac{2.5+j1.5}{22.5-j12.5}(100) = \frac{2.915\angle 30.96°}{25.74\angle -29.05°}(100) = 11.32\angle 60.01°$$

$$\mathbf{V}_2 = \frac{7.5+j4}{22.5-j12.5}(100) = \frac{8.5\angle 28.07°}{25.74\angle -29.05°}(100) = 33.02\angle 57.12°$$

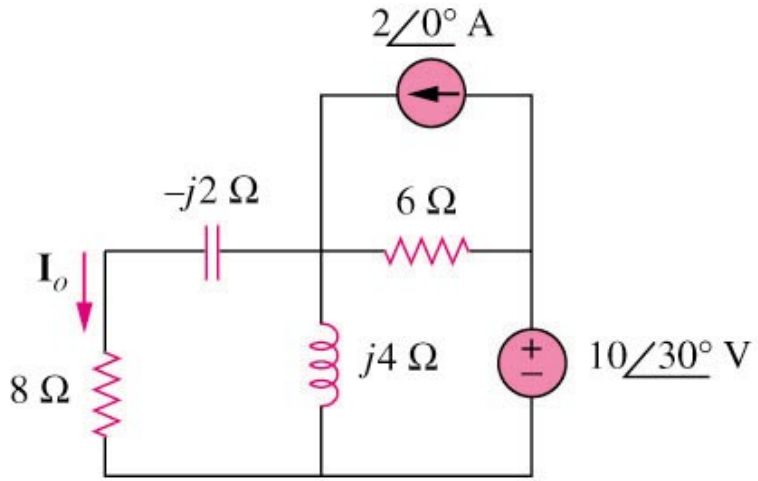
✓ In time domain using sin convention,

$$v_1(t) = \mathbf{11.32\ \sin(2t+60.01°)\ V}$$
$$v_2(t) = \mathbf{33.02\ \sin(2t+57.12°)\ V}$$

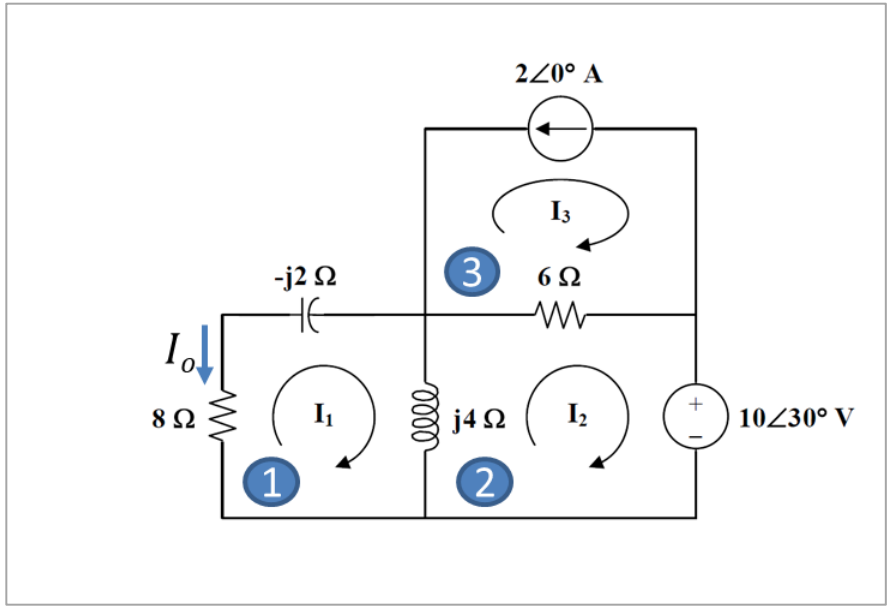
Mesh Current Analysis

✓ Mesh analysis uses Kirchhoff's Voltage Law (KVL) to determine the currents in each mesh (loop) of a circuit.

❑ Find I_o for the following circuit using mesh analysis.



I_1, I_2, I_3 be the mesh currents in the circuit



For mesh 1, KVL provides: $(8 - j2 + j4)\mathbf{I}_1 - j4\mathbf{I}_2 = 0$

For mesh **2**, KVL provides: $(6 + j4)\mathbf{I}_2 - j4\mathbf{I}_1 - 6\mathbf{I}_3 + 10\angle 30° = 0$

For mesh **3**: $\mathbf{I}_3 = -2$ Note: $\mathbf{I}_o = -\mathbf{I}_1$

By solving the above three equations we obtain: $\mathbf{I}_o = \underline{\mathbf{1.194\angle 65.44°\ A}}$

The Principle of superposition

Step1: Retain one independent source at a time in the circuit and replace all other sources with their internal impedances.

Voltage source → short circuit

Current source → open circuit

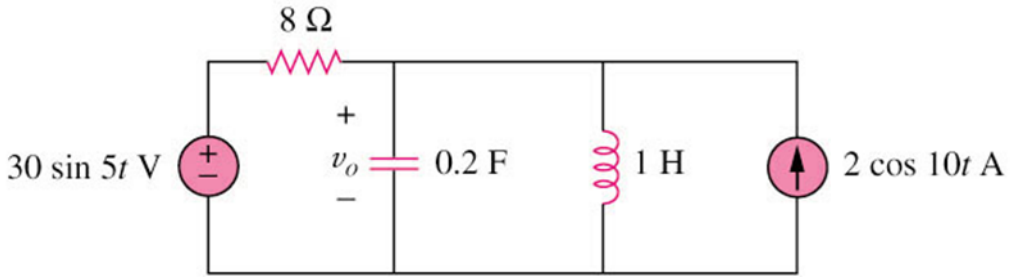
Step2: Determine the output (current or voltage) due to the single source acting alone using any circuit analysis methods.

Step3: Repeat steps 1 and 2 for each of the other independent sources.

Step4: Find the total contribution by adding algebraically all the contributions due to the independent sources.

Example 18

- Calculate v_o for the circuit shown below using the superposition theorem.



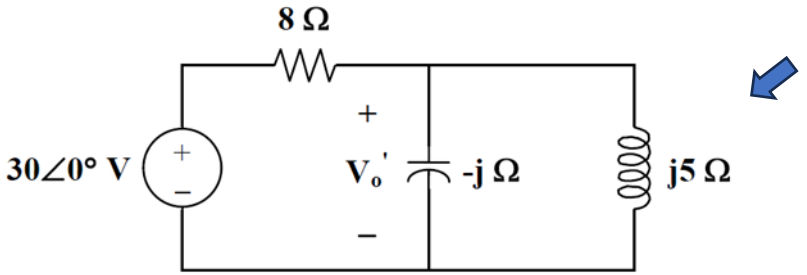
Solution

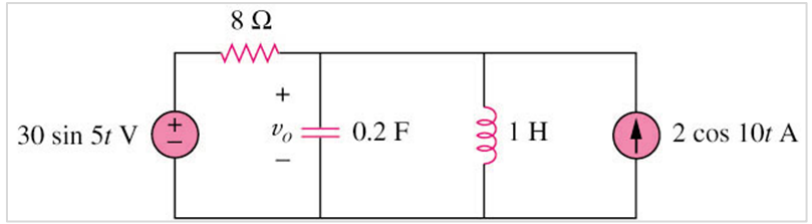
- ✓ Let $v_o = v_o' + v_o''$
- ✓ v_o' is due to the voltage source
- ✓ v_o'' is due to the current source.
- ✓ Note that frequencies of voltage and current sources are not the same.

Superposition Theorem Example

- To calculate v_o', we remove the current source (open circuit)

$$30\sin(5t) \longrightarrow 30\angle 0°, \quad \omega = 5$$
$$0.2\text{ F} \longrightarrow \frac{1}{j\omega C} = \frac{1}{j(5)(0.2)} = -j$$
$$1\text{ H} \longrightarrow j\omega L = j(5)(1) = j5$$

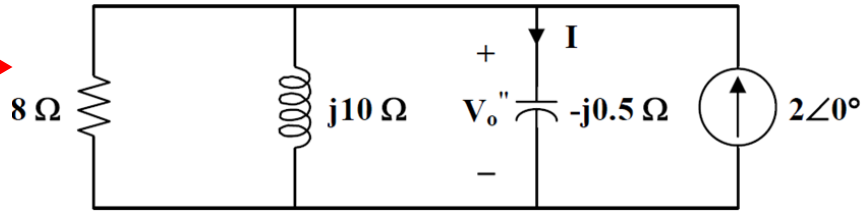




- By voltage division $\mathbf{V}_o' = \dfrac{-j1.25}{8 - j1.25}(30) = 4.631\angle -81.12°$ { $-j \| j5 = -j1.25$ }; $v_o' = 4.631\sin(5t - 81.12°)$

- To calculate v_o'', we short circuit the voltage source

$$2\cos(10t) \longrightarrow 2\angle 0°, \quad \omega = 10$$
$$0.2\text{ F} \longrightarrow \frac{1}{j\omega C} = \frac{1}{j(10)(0.2)} = -j0.5$$
$$1\text{ H} \longrightarrow j\omega L = j(10)(1) = j10$$



$$\mathbf{Z}_1 = -j0.5,$$
$$\mathbf{Z}_2 = 8 \| j10 = \frac{j80}{8 + j10} = 4.878 + j3.9$$

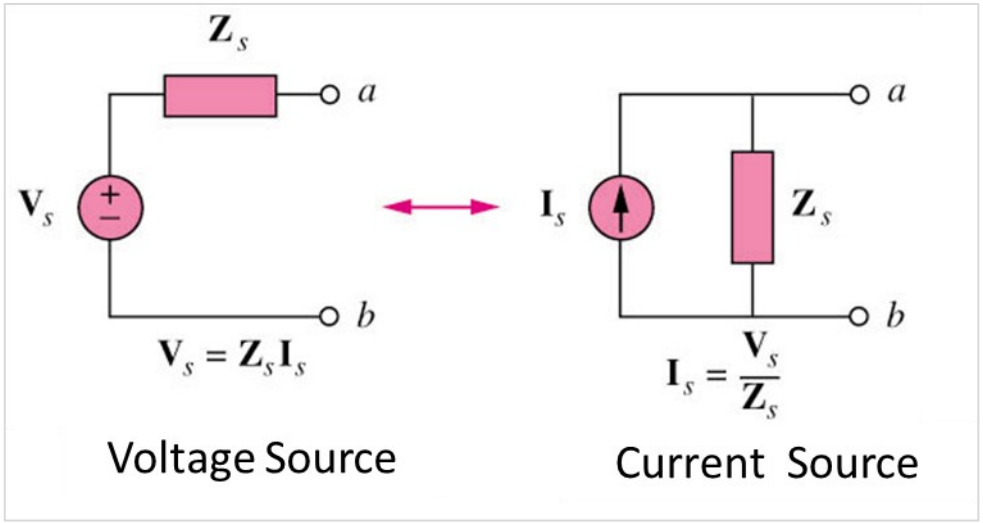
$$\mathbf{V}_o'' = \mathbf{I}(-j0.5) = \frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2}(2)(-j0.5) = \frac{-j(4.877 + j3.9)}{4.878 + j3.4}$$
$$\mathbf{V}_o'' = \frac{6.245\angle -51.36°}{5.94\angle 34.88°} = 1.051\angle -86.24°$$
$$v_o'' = 1.051\cos(10t - 86.24°)$$

Add all contributions algebraically:

$$v_o = v_o' + v_o''$$

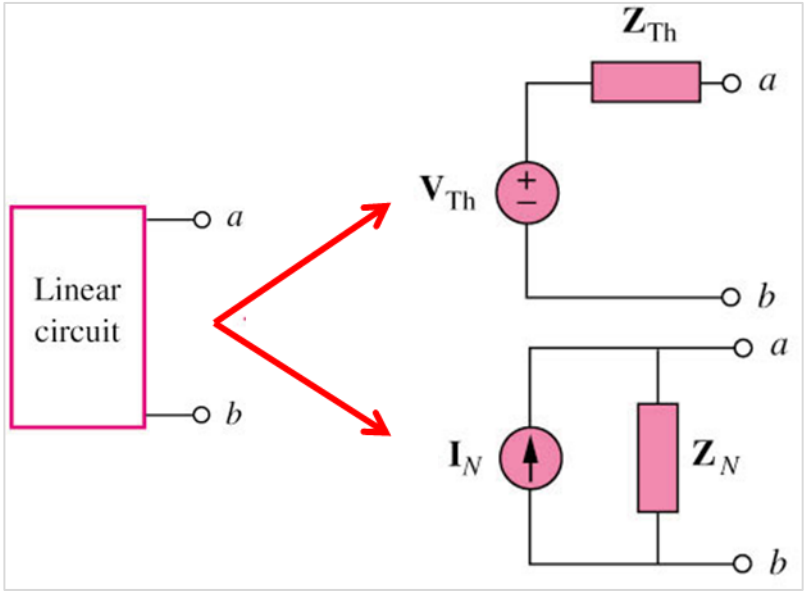
v_o = **4.631 sin(5t – 81.12°) + 1.051 cos(10t – 86.24°) V**

Transformation between voltage and current Sources



Note: The Internal resistance of the voltage source has the same value as the internal resistance of the current source.

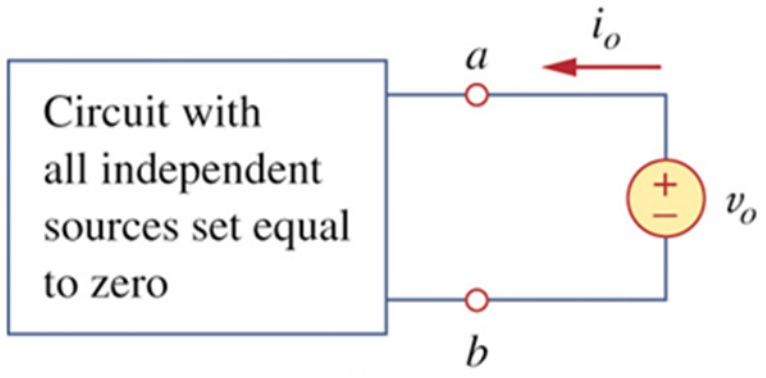
Thevenin's and Norton's Equivalent Circuits



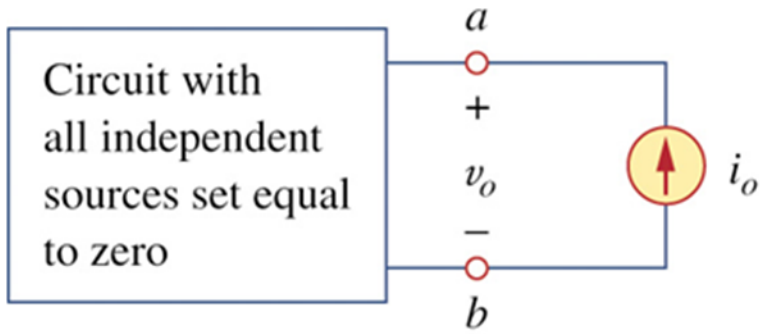
- ✓ Thevenin's and Norton's theorems are applied to ac circuits in the same way as they are to dc circuits.
- ✓ Always disconnect the load impedance (Z_L) from the circuit (between *a* and *b*) before finding the Thevenin's or Norton's equivalent circuit

Calculating Z_{Th} for circuits with dependent sources

- Remove all the independent sources.
- The dependent sources cannot be removed.
- Assume a voltage or current source connected across a-b terminals as shown.
- The value assumed is generally $1\angle 0°$.
- Calculate the voltage or current (v_o or i_o) and hence $Z_{Th} = \frac{v_o}{i_o}$



(a)

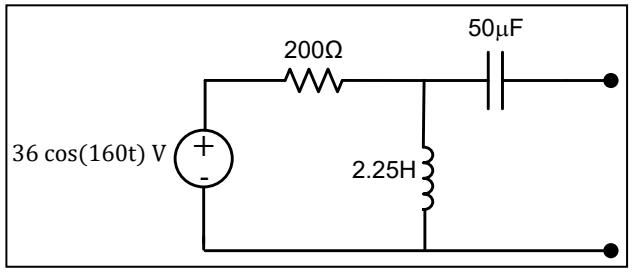


(b)

Examples

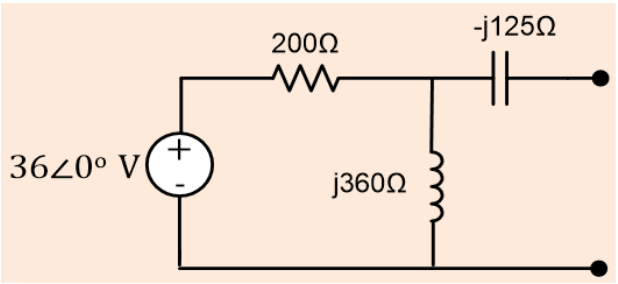
Example 19

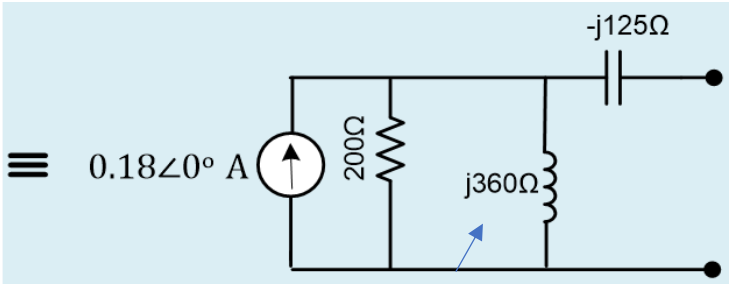
❑ Use **source transformation** and equivalent impedance to find Thevenin equivalent circuit of the ac circuit shown on the right

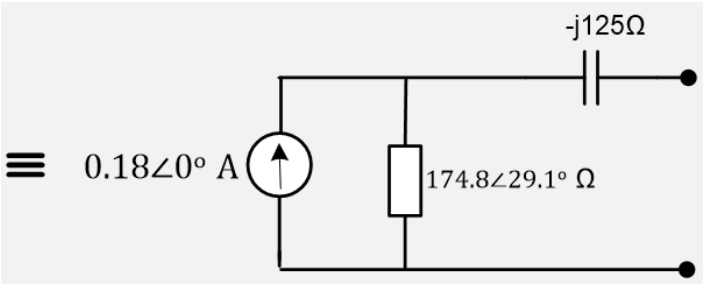


Solution

- It can be seen that the above circuit contains a series combination of a voltage source and impedance
- A source transformation replaces this series combination with a parallel combination of a current source and impedance as shown below:

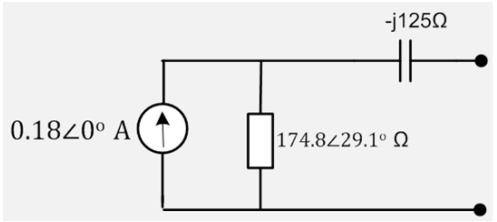


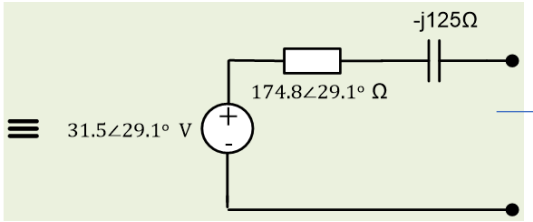


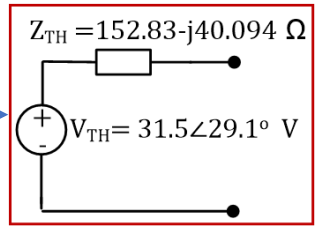


$$200 \| j360 = \frac{200(j360)}{200+j360} = 174.8\angle 29.1^{o}$$

- The impedances of the resistor and inductor are connected in parallel, and we can calculate the equivalent impedance.
- A source transformation replaces this parallel combination with a series combination of a voltage source and impedance. We can calculate the equivalent impedance: $174.8\angle 29.1^{o} - j\,125 \;=\; 152.83 \;+\; j\,84.907 - j125 \;=\; 152.83 - j40.094$





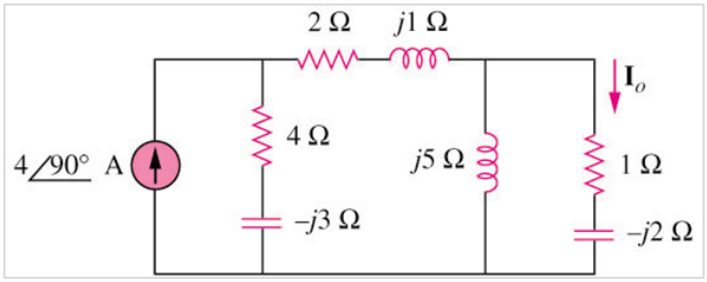


Thevenin equivalent circuit is:

Examples

Example 20

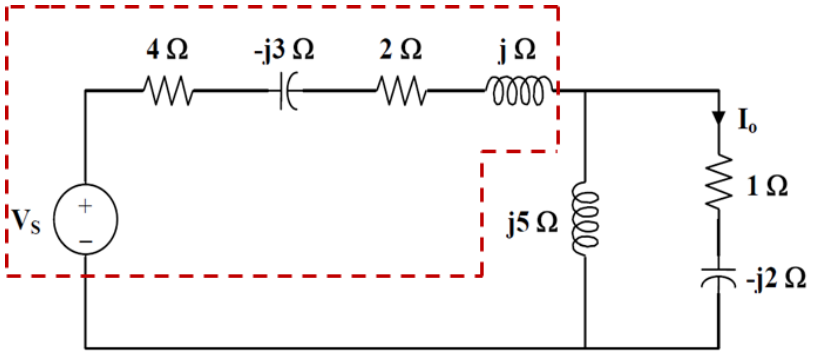
- Find I_o for the circuit shown below using source transformation.



Solution

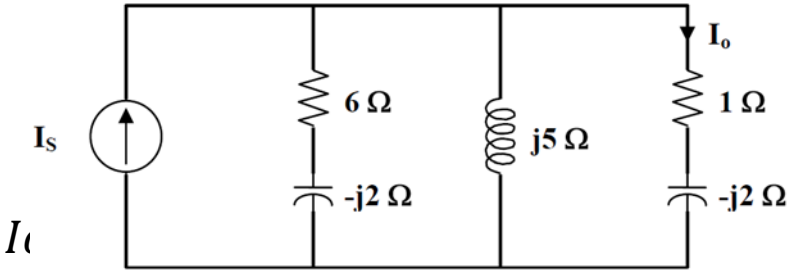
Step1: Let's transform the current source to a voltage source:

$$\boldsymbol{I_S} = j4A, \quad Z_s = 4 - j3 \qquad \mathbf{V}_s = \mathbf{I}_s \mathbf{Z}_s = (\mathrm{j}4)(4 - \mathrm{j}3) = 12 + \mathrm{j}16$$



Step 2: Now convert the voltage source ,with new impedance Z ,(shown by the dotted line) to a current source.

$$\mathbf{Z} = 4 - \mathrm{j}3 + 2 + \mathrm{j} = 6 - \mathrm{j}2 \qquad \mathbf{I}_s = \frac{\mathbf{V}_s}{\mathbf{Z}} = \frac{12 + \mathrm{j}16}{6 - \mathrm{j}2} = 1.5 + \mathrm{j}3$$



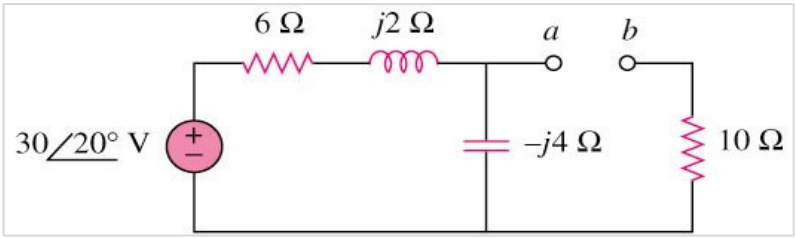
Find equivalent impedance of (6-j2) & j5. and then apply current division to find I_o

Ans: $I_o = 3.288\angle 99.46^o$ A

Examples

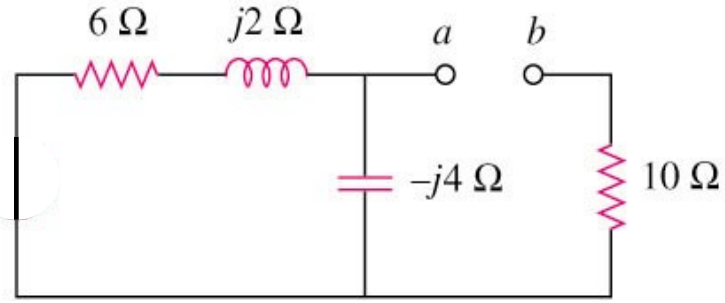
Example 21

- Find the Thevenin's equivalent between terminals *a* and *b* for the circuit shown on the right.



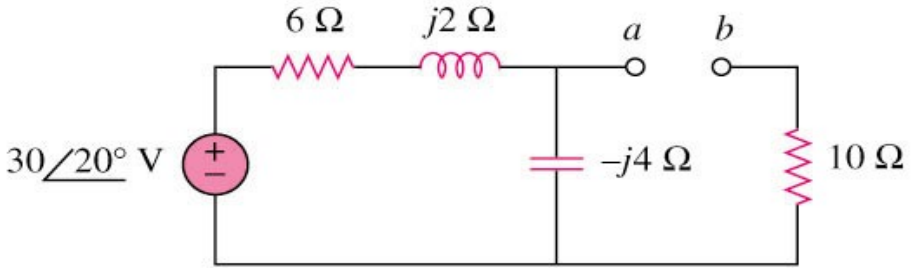
Solution

Step1: The voltage source is turned off by setting it to zero (i.e short circuit).



$$Z_{TH} = 10 + (-j4) \,||\, (6 + j2)$$
$$= 10 + \frac{(-j4)(6 + j2)}{6 - j2}$$
$$= 10 + 2.4 - j3.2$$
$$= \underline{\mathbf{12.4 - j3.2\ \Omega}}$$

Step1: V_{TH} is open circuit voltage at terminals *a* and *b* with the voltage source turned on.



$$V_{TH} = \frac{-j4}{6 + j2 - j4}(30\angle 20°) = \frac{(-j4)(30\angle 20°)}{6 - j2}$$
$$= \frac{(4\angle -90°)(30\angle 20°)}{6.324\angle -18.43°}$$
$$= \underline{\mathbf{18.97\angle -51.57°\ V}}$$

ELEC2134: Circuits and Signals

AC Circuit Analysis

Tutorial questions

AC Circuit Theorems

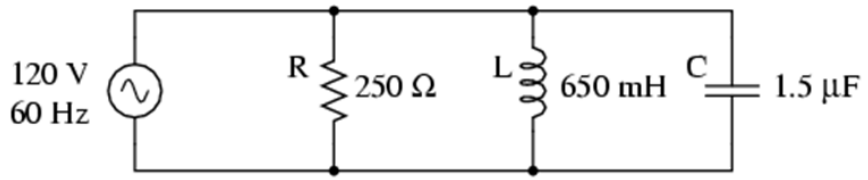
- ✓ Impedance (Z), Admittance (Y), Conductance (G) and Susceptance (B)
- ✓ Delta(Δ) - Wye (Y) Transformation in AC networks
- ✓ Independent and dependent current and voltage sources
- ✓ Nodal Analysis
- ✓ Mesh Analysis
- ✓ Superposition Theorem
- ✓ Thevenin's and Norton's Equivalent Circuits

Adapted by Aous Naman (PhD) from lectures by
Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

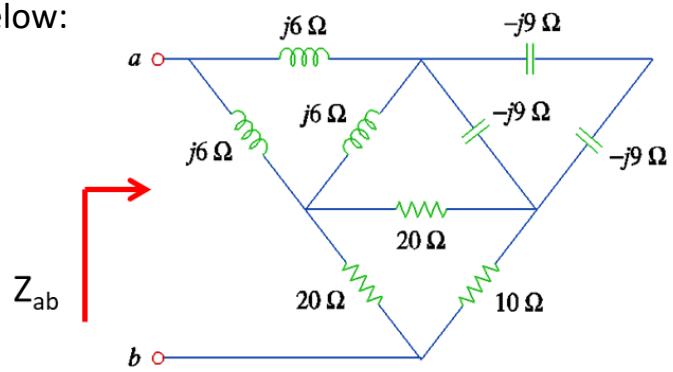
Tutorial Questions (AC Circuit Theorems)

Question 1: Determine the total impedance of the circuit below using the admittance method.



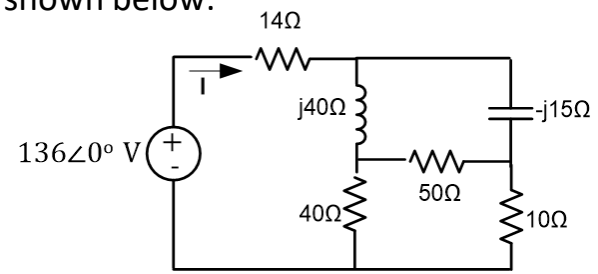
Ans: $188.68\angle 41.2^{o}$

Question 2: Calculate the impedance (Z_{ab}) for the network shown below:



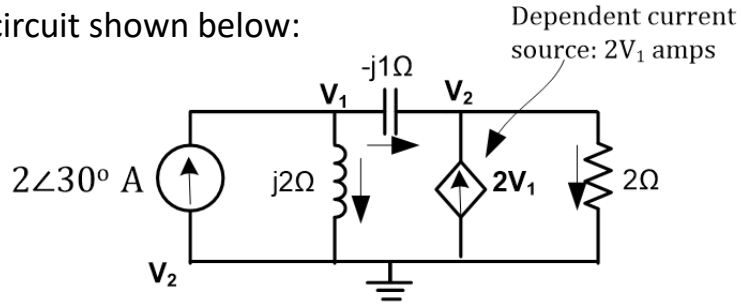
Ans: $7.567 + j0.5946\ \Omega$

Question 3: Use a Δ - to - Y transformation to find the current **I** in the circuit shown below:



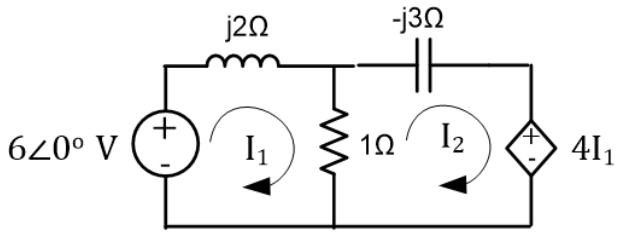
Ans: $\mathbf{I} = 4\angle 28.07^{o}$ A

Question 4: Find V1 and V2 via nodal analysis for the circuit shown below:



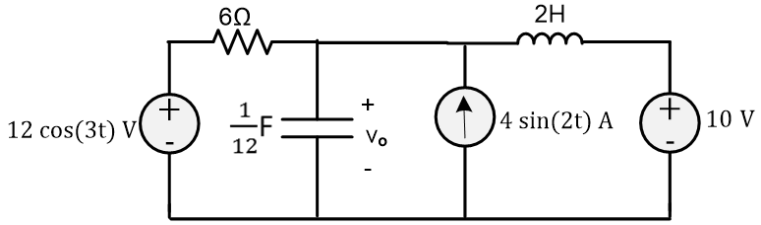
Ans: $\mathbf{V}_1 = 1.229\angle 167.49^{o}$ V ; $\mathbf{V}_2 = 2.457\angle 130.62^{o}$ V

Question 5: Find I_1 and I_2 via mesh analysis:



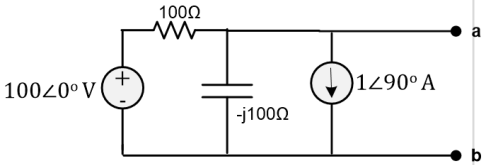
Ans: $\boldsymbol{I_1} = 2.095\angle -65.23^{o}$ A; $\boldsymbol{I_2} = 1.988\angle -173.66^{o}$ A

Question 6: Solve for v_0 for the circuit shown below using the superposition principle.



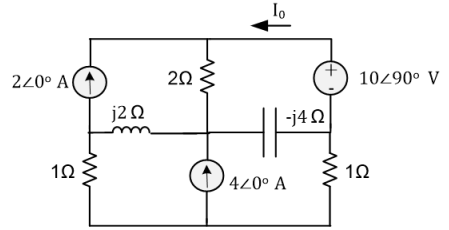
Ans: $v_o = 10 + 21.45\sin(2t + 26.56^{o}) + 10.73\cos(3t - 26.56^{o})$ V

Question 7: Find the Thevenin and Norton equivalent circuits, for the circuit shown below, between terminals a and b:



Ans: $\boldsymbol{Z_N} = \boldsymbol{Z_{TH}} = 50 - j50\boldsymbol{\Omega}$,
$\boldsymbol{V_{TH}} = 100\angle 0^{o}$ V,
$\boldsymbol{I_N} = 1.414\angle 45^{o}$ A

Question 8: Using mesh analysis, obtain $\mathbf{I_0}$ for the circuit shown below



Ans: $3.35\angle 174.3^{o}$ A